

PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience in full-stack development, backend systems, and AI/ML applications using Python, natural language processing, and data structures. Currently building production features at CharityAuctions using React/Next.js, Node.js/Python, and REST APIs, collaborating with cross-functional teams and product management to deliver scalable solutions. Graduate Teaching Assistant – Artificial Intelligence at UNC Charlotte, supporting instruction on machine learning, generative AI, and user-centered design principles.

TECHNICAL SKILLS

AI /ML & NLP: Natural Language Processing, Machine Learning, Generative AI, Neural Networks, Model Training, PyTorch, TensorFlow, Hugging Face

Programming Languages: Python, JavaScript (ES6+), TypeScript, PHP, SQL, Java, C++

Data Structures & Algorithms: Trees, Graphs, Hash Maps, Heaps, Dynamic Programming, Sorting, Searching, Algorithm Optimization
Full-Stack Development: React, Next.js, Angular, Node.js, Express.js, Django, Flask, REST APIs, Microservices

Frontend & UI: UI Design, HTML5, CSS3, Tailwind CSS, Bootstrap, User-Centered Design, Responsive Design

Databases: MySQL, PostgreSQL, MongoDB | **Cloud & DevOps:** AWS, Azure, Docker, Git, CI/CD, GitHub Actions

Product & Collaboration: Product Management collaboration, Cross-Functional teams, Agile/Scrum, Sprint Planning, PRD Documentation

EXPERIENCE

Software Engineer Intern | CharityAuctions (Chicago, IL) (charityauctions.com) **Jul 2025 – Present**

- Built production features for Auction 2.0 platform using React/Next.js, TypeScript, Node.js, and **Python**, implementing optimized **data structures** for auction listing workflows that improved platform performance and increased mobile engagement by 40%
- Developed reusable **UI design** components including QR-code item previews and responsive auction flows, applying **user-centered design** principles validated through user testing that enhanced user experience and bidding activity
- Enhanced backend workflows with **Python** and efficient **data structures** for data processing, integrating REST APIs with PostgreSQL databases to improve system reliability and reduce API latency by 25%
- Fixed production bugs across auction and bidding flows, implementing algorithm optimizations and debugging techniques that improved responsiveness and reduced support tickets by 30%
- Collaborated with **product management** and engineering teams in Agile sprints to scope roadmap features, automate nonprofit onboarding workflows, and deliver **full-stack** solutions that improved release stability and delivery timelines by 20%
- Participated in **cross-functional** code reviews, sprint planning, and issue tracking alongside engineering, design, and QA teams, improving overall development turnaround time by 15% through clearer documentation and testing workflows

Software Developer Intern | Creative Central (India) (creativecentral.in) **Jun 2022 – Jun 2024**

- Engineered **full-stack** applications using React.js, Angular, TypeScript, **Python** (Django, Flask), and PHP, leveraging optimized **data structures** and algorithms that boosted backend performance and UI rendering speed by 20-25%
- Architected and optimized RESTful APIs using **Python** Flask, Node.js, and Express.js, integrating PostgreSQL and MongoDB databases with advanced indexing strategies (B-trees, hash indexes) for scalable data persistence
- Partnered with **cross-functional** teams and **product management** to translate business requirements into technical solutions, applying **user-centered design** methodology validated through user testing
- Deployed CI/CD pipelines using GitHub Actions and Docker containerization for automated deployments on AWS and Azure, reducing deployment failures by 30% and streamlining release cycles
- Designed responsive **UI** components using HTML5, CSS3, Bootstrap, and Tailwind CSS, ensuring cross-browser compatibility and **user-centered** experiences across Chrome, Firefox, Safari, and Edge

LEADERSHIP & TEACHING EXPERIENCE

Graduate Teaching Assistant – Artificial Intelligence | University of North Carolina at Charlotte **Aug 2025 – Dec 2025**

- Supported graduate AI course (ITCS 6150/8150) for 75+ students covering machine learning, **natural language processing**, **generative AI**, and **data structures** for neural networks, assisting with search algorithms, heuristics, and AI problem-solving using **Python**
- Evaluated student projects on machine learning models and **natural language processing** applications, providing debugging support and code review feedback on algorithm efficiency, **data structures** optimization, and **user-centered design** principles
- Led office hours and coordinated with instructor on assignment design, grading standards, and project guidance while enforcing academic integrity and best practices
- Assisted students with projects involving **LLMs and transformer-based models**, providing guidance on **prompt design**, **model evaluation**, and **generative AI best practices** using Python and Hugging Face

- Event Organizer – TEDxGPREC
- Jun 2022 – Dec 2023
- Coordinated large-scale TEDx event for 750+ attendees, increasing registrations and online engagement by 40% through **cross-functional** collaboration and stakeholder coordination, applying **product management** principles for task prioritization and workflow optimization

PROJECTS

- AI-Powered Chatbot using GPT API | Python, GPT API, NLP, Full-stack
- Built an **AI-powered conversational chatbot** using the **GPT API**, implementing **prompt engineering**, **context management**, and **response validation** to ensure accurate, coherent, and safe natural language interactions
 - Implemented **Python backend services** for **LLM inference**, **conversation state handling**, **API rate limiting**, and **automated evaluation** of response quality, factual consistency, and latency, integrating the chatbot into a **full-stack web application** with a responsive UI
- GVHMR (Generative Video Human Mesh Recovery) | Python, PyTorch, Hugging Face
- Implemented **end-to-end deep learning pipeline** using **Python**, **PyTorch**, and **Hugging Face Transformers**, leveraging advanced **data structures** (spatial graphs, efficient queues) for **3D human pose estimation**, reducing **inference time by 35%**
 - Trained and optimized **Transformer-based neural network models** using **PyTorch** with custom **batch-processing data structures**, building **automated evaluation pipelines** with visualization tools following **user-centered design principles**
 - Utilized **Hugging Face Transformers** for **model training and inference**, developing **evaluation workflows** to assess **output consistency, accuracy, and performance**, aligned with **modern generative AI system practices**
- Health-Focused Wellness Tracking Application | Python, NLP, Full-stack
- Developed health tracking application using **Python** and **natural language processing** for wellness data analysis, building **full-stack** web app with React frontend and Flask backend with intuitive **UI design** components
 - Implemented efficient **data structures** (time-series trees, hash maps) for health metrics storage and real-time analytics, collaborating with **product management** stakeholders to define features aligned with user needs
- ChronoPulse – Real-Time Analytics Dashboard | Python, Full-stack
- Built scalable **full-stack** application using **Python** Flask and React, implementing **data structures** (time-series trees, hash maps) for real-time analytics processing and SQL storage
 - Designed responsive **UI** following **user-centered design** methodology, collaborating with **product management** to define requirements and prioritize features
- QR-Based Attendance System with GPS Verification | Python, Full-stack
- Developed **full-stack** attendance system using **Python** with efficient **data structures** (geospatial hash maps, bloom filters) for real-time GPS validation and spoofing prevention, ensuring attendance accuracy
 - Designed **user-centered** interface using React with responsive **UI design**, conducting user testing with 100+ students to optimize accessibility and ease of use

EDUCATION

- Master of Science in Computer Science, University of North Carolina at Charlotte (GPA: 4.0)
- May 2026
- Bachelor of Technology in Computer Science & Engineering, G. Pulla Reddy Engineering College (GPA: 8.05)
- June 2024

CERTIFICATION

- AWS Academy Graduate - AWS Academy Machine Learning Foundations
- Python Programming Certification – HackerRank
- AWS Academy Graduate - AWS Academy Data Engineering
- AI/ML Virtual Internship (AICTE)
- Data Engineer Virtual Internship